

Technical Specification & Architecture

PWD BUPCA Inc. Platform Specification

Enterprise & Community Operational Command Center • Accessible Public Store • Station DTR & Payroll Engine

Client / Community: PWD BUPCA Inc. (Brgy. UP Campus, Diliman)

Application Framework: Next.js 16 (App Router, Turbopack, TypeScript)

Database Layer: Supabase (PostgreSQL 15 + PostGIS + Realtime)

Recognition: UP Gawad Tsanselor 2025 Awardee

Styling & Design System: Tailwind CSS v4 (@custom-variant dark)

Typography: Plus Jakarta Sans & Newsreader Serif

1. Platform Architecture & Modules Overview

The platform features a dual-facing modular architecture engineered with universal design principles (WCAG 2.1 AA/AAA compliance):

- Public Web Portal** (`/``, `/store``, `/programs``): Brand storytelling, certified livelihood program enrollment, and a fully accessible e-commerce virtual store supporting GCash, BPI UP Campus bank transfer, and on-campus pickup.
- Command Center** (`/admin``): A responsive sidebar-driven operations suite integrating GIS asset tracking, 100% subsidized raw materials ordering, order fulfillment pipeline, dynamic mathematical piece-rate payroll, and a no-code Website Content Manager (CMS).

2. Universal Accessibility Engineering (WCAG 2.1 AA/AAA)

Accessibility Feature	Implementation Technique	User Benefit
Theme Switcher	Tailwind CSS v4@custom-variant dark with local storage caching	Prevents glare and eye strain with seamless light and dark palettes
High Contrast Mode	Yellow text on pitch-black background (4.5:1 to 7:1 ratio)	Essential for low-vision and macular degeneration users
Dynamic Font Scaling	Dynamic 100% – 150% text size scaling without layout clipping	Allows comfortable reading for seniors and visually impaired members
Dyslexia Friendly Font	OpenDyslexic typeface with weighted bottoms and wide kerning	Significantly reduces letter confusion and inversion
Sensory Audio Cues	Web Audio API oscillator chimes & SpeechSynthesis voice feedback	Provides auditory confirmations for clock-ins, orders, and CRUD saves

3. Database Schema & Entities (PostgreSQL / Supabase)

Full schema deployed via supabase_schema.sql including foreign keys, cascades, and enum classifications:

Database Table	Primary Keys & References	Key Columns & Data Types
profiles	id (UUID, PK) references auth.users	full_name, role (superuser/admin/member_operator), disability, phone
locations	id (UUID, PK)	name, address, latitude, longitude, contact_person
assets	id (UUID, PK), location_id (FK)	asset_tag (BUPCA-SEW-01), category, status (online/idle/maintenance), specs
asset_logs (DTR)	id (UUID, PK), asset_id (FK), member_id (FK)	clock_in, clock_out, hours_worked, units_produced, materials_consumed (JSONB)
supplies	id (UUID, PK)	item_name, unit (meters/pieces), current_stock, min_threshold, is_subsidized
products	id (UUID, PK)	title, price, stock, artisan_attribution, default_piece_rate, featured
orders & order_items	id (UUID, PK), product_id (FK)	order_number, customer_name, total_amount, payment_method, status
payroll_records	id (UUID, PK), member_id (FK)	hours_worked, hourly_allowance, product_breakdown (JSONB), gross_pay, net_payout

4. Command Center Operations & Mathematical Engine

4.1. Asset Operations & Interactive GIS Map

Integrates Leaflet and OpenStreetMap centered on UP Diliman coordinates (14.6548° N, 121.0682° E) linking machinery to physical nodes: UP CHE Workshop, Area 2 Community Center, and Pook Dagohoy Headquarters. Operators perform digital Daily Time Record (DTR) clock-ins with real-time relocation logs.

4.2. Mathematical Payroll Engine & Piece-Rate Multipliers

The payroll ledger automatically itemizes varied product lines produced during a pay period:

Hourly Allowance = Monitored DTR Hours × ₱85.00/hr

Total Piece-Rate Earnings = Σ (Units Produced[i] × Piece Rate[i])

Gross Pay = Hourly Allowance + Total Piece-Rate Earnings

Net Artisan Payout = Gross Pay + Adjustments (0% Material Deduction – 100% Grant Covered)

Product Name	Product Category	Retail Price	Labor Piece-Rate
BUPCA Signature Denim Tote Bag	Upcycled Eco-Bags	₱380.00	₱75.00 / unit
Foldable Fabric Bread Basket	Fabric Baskets	₱250.00	₱55.00 / unit

Product Name	Product Category	Retail Price	Labor Piece-Rate
Compact Denim Utility Pouch	Denim Pouches	₱195.00	₱45.00 / unit
Community Patchwork Canvas Apron	Upcycled Home Crafts	₱420.00	₱85.00 / unit

4.3. Website Content Manager (CMS with Full CRUD)

Includes 3 operational tabs:

- **Landing Page & Narrative:** Edit headlines, serif highlight text, Gawad Tsanselor notice, and verified impact metrics.
- **Livelihood Programs (Full CRUD):** Manage curricula, UP academic partners, duration, cohort capacities, and deliverables on /programs.
- **Artisan Store (Full CRUD):** Manage retail prices, stock quotas, product stories, photos, and labor piece-rates on /store.

5. Deployment & Execution Instructions

```
# 1. Clone or navigate to the Next.js directory cd pwd-bupca-app # 2. Install dependencies npm install # 3. Start local development server npm run dev # Live at: http://localhost:3000 (Public) and http://localhost:3000/admin (Command Center) # 4. Compile optimized production build npm run build
```